

Raffay Hassan

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Portfolio: <https://raffayhassan-portfolio.netlify.app/>

PERSONAL STATEMENT

"I am a dedicated final year Computer Systems Engineering student (expected First-Class Honours) with expertise in building real-world digital and embedded systems using C++ and Python. I have practical experience in AI/Autonomous Driving research (LiDAR-camera fusion) and designing IoT systems, which was shortlisted for industrial showcase. I am proficient in tools like Xilinx Vivado, MATLAB, and SolidWorks, and I consistently apply Agile methodologies and strong problem-solving skills to technical challenges."

EDUCATION

Middlesex University, London, UK bachelor's in **computer systems engineering** (Expected Graduation: 2026)

- **Achieved First-Class grades** (Expected to Graduate in **First class honours**)
- Relevant Coursework: Software Engineering, Digital Systems, Machine Learning, Cybersecurity, Signal Processing, Programming Paradigms, Cloud Computing

PROFESSIONAL EXPERIENCE

Sensor-Driven Digital Twin for Collision Prevention in Autonomous Systems [2026]

Final Year Dissertation Project

- Designed and implemented a real-time multi-sensor system integrating lidar, radar, and camera for predictive collision detection on embedded hardware
- Developed a digital twin model to represent the vehicle environment and enable time-to-collision based risk assessment
- Built and deployed the system across simulation (CARLA) and physical hardware platforms
- Implemented sensor fusion using a safety-driven worst-case decision strategy for robust collision avoidance
- Achieved reliable obstacle detection and real-time performance in controlled testing
- Worked with NVIDIA Jetson, Raspberry Pi, and Arduino for system integration
- Applied YOLO-based computer vision and real-time processing using Python and C++

Intern at London Digital Twin Research Centre [2025]

- **Autonomous Driving Research:** Contributed to object detection and **LiDAR-camera data fusion** using IDD Multimodal and IDD-Supplement datasets.
- **AI/Simulation Tools:** Explored object detection pipelines (**YOLO** and **PointPillars**), initiating inference **benchmarking** research, and developed traffic route simulations in **SUMO** integrated with **Unity**.
- **Data & Calibration:** Designed and implemented a **LiDAR-camera fusion strategy** for annotating unlabelled data, and built a custom **Python UI calibration tool** for frame alignment.
- **Dataset Validation:** Audited large-scale datasets (IDD Detection) using **Python scripts** to validate file integrity and delivered documented methodologies for future annotation workflows.

IoT-Based Indoor Air Quality Monitoring System [2025]

- Siemens Connected Curriculum Showcase project
- Developed an IoT system using ESP8266, DHT22, and SGP30 sensors to monitor indoor air quality.
- Integrated cloud storage and Mendix for real-time dashboards and alerts.

- Demonstrated skills in embedded C++, cloud APIs, low-code, and OAuth2 setup for Siemens Insights Hub.

Arduino Projects Developer [2023] University Project.

- Developed multiple mini projects using **Arduino**, integrating **sensors** and creating real-world applications with **Python** and **C++**.
- Utilized **GitHub** for version control and collaborated effectively with team members.
- Gained hands-on experience working with hardware, software integration, and real-time systems.

Mechanical Design & CAD Modelling [2023] University Project.

- Utilized **SolidWorks** to create 3D models, simulations, and technical drawings for product designs.
- Optimized designs for manufacturability, improving project efficiency.

Signal Processing [2024] University Project.

- Performed **signal processing** tasks using **MATLAB**, including data filtering, transformation, and analysis for project applications.

FPGA Design and Development [2024]

Designed and developed FPGA circuits using **Xilinx Vivado**, focusing on **digital logic design** and circuit optimization.

COMPETITIONS AND ACHIEVEMENTS

- Shortlisted for Siemens Connected Curriculum Industrial Showcase 2025
- Reached National Qualifier stage – WorldSkills UK 2025 (Industrial Electronics)

EXTRA COURSES

- **Intermediate Ethical Hacking and Cybersecurity** (Udemy, in progress).
- **Unreal Engine 5 Game Development (C++ & Blueprint)** (Udemy, in progress).

CORE SKILLS

- **Programming Languages:** C++, Python, Unreal Engine 5 (C++ and Blueprint) , VHDL
- **Cybersecurity:** Ethical Hacking, Vulnerability Analysis, System Protection Techniques
- **Tools & Technologies:** Arduino, Cloud Computing, AI.
- **Software Tools:** SolidWorks (Mechanical Design Projects), Xilinx Vivado (FPGA Development), Altium Designer (PCB Design), MATLAB (Simulation & Modelling), Multisim (Circuit Simulation), Code Composer Studio (Embedded Development) Mendix (low-code app development) , Siemens insights hub.
- **Problem Solving:** Debugging, Performance Optimization
- **Collaboration & Leadership:** Group Leadership, Peer Mentorship, Cross-Functional Teamwork
- **LiDAR-Camera Data Fusion**
- **Object Detection** Pipelines (YOLO, PointPillars)
- **Dataset Auditing & Validation**
- **Autonomous Driving Simulation** (SUMO, Unity Integration)

ADDITIONAL INFORMATION

- **Languages:** English (Fluent) , Urdu (Fluent).
 - **Interests:** AI, Cybersecurity, Cloud Computing, Software Engineering, Game Development, Research
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